

**Construction and Validation of the Cyber Psychological Security:
a survey study on Omani university students**

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Construction and Validation of the Cyber Psychological Security: a survey study on Omani university students

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Abstract: This study develops the psychological security scale of Omani university students in cyberspace. A sample of 110 participants from Dhofar and Sultan Qaboos Universities (40% male, 60% female). Participants were categorized according to their daily social media usage: 52.7% used social media for less than three hours per day, while 47.3% used it for more than three hours. A descriptive approach was employed to assess the validity of the study instrument. Cyber psychological security was measured using a 42-item scale across seven subscales: emotional well-being, personal independence, confidence, credibility, privacy and anonymity, social support, and sense of power. Reliability coefficients ranged ($\alpha = .710$ to $.949$, $\omega = .731$ to $.950$), indicating acceptable internal consistency. Descriptive statistics revealed moderate to high psychological security levels, with personal independence and credibility scoring highest and sense of power lowest. Skewness and kurtosis indices

confirmed normal distribution. Exploratory factor analysis identified three primary factors, explaining 57.6% of the variance ($KMO = .87$). A second-order analysis validated a hierarchical model, with all seven subscales loading onto a general psychological security factor ($KMO = .896$; explained variance = 85.2%). The findings underscore the cultural significance of personal independence and trustworthiness in Omani society while highlighting areas for enhancing digital empowerment.

Keywords: Psychological security; Cyberspace; Cybersecurity.

Introduction

Cybersecurity in Omani society serves as a strategic pillar for safeguarding national interests, driving sustainable economic advancement, and preserving social harmony and resilience (Ahmad et al., 2024). Psychological security is defined as a feeling that appears because of one's sense of danger, physiologically or psychologically, or it is a feeling of helplessness and powerlessness that one has during dealings with others. It is a feeling of certainty and the ability to control oneself, and it is a personal judgment on whether the individual's environment can be controlled (Wang et al., 2019), and this depends on personality traits and ego strength. Al Diyar & Salem's (2015) study demonstrated a positive relationship between achieving ego identity and psychological security. Security is considered the basis of response to the fear of crime or social security and is linked to some social factors, such as social awareness. Psychological security is divided into three categories: psychological, social, and environmental (Wang et al., 2019). The function of the ego is linked to the organization of personality. The functional relationship between the ego, personality, and self causes continuity and interaction, especially virtual ones, in pursuit of the ideal to compensate for the weakness and shortcomings of the ego in live interactions (Shin, 2010). Psychological security plays a role in controlling the ego during virtual interactions to reduce impulsiveness and achieve personal flexibility in a dynamic way to control oneself in situational situations, which leads to emotional satisfaction and social compensation and reduces problems and mood disorders to achieve harmony (Letzring et al., 2005). Ideal and realistic virtual interactions of the self lead to the formation of unique psychological perceptions and an increased desire for emotional stability (Perlin & Li, 2020; Shin, 2010). From another perspective, there is a fear on the part of individuals of falling victim to security threats or phishing by publishing posts and comments on their walls (Molodetska et al., 2020). Psychological security may be a fear of falling victim to the dark trinity of exposure and harassment by the narcissist, psychopath, and Machiavellianism, as the harasser's motive may be silence. It is the disappearance of a prior reaction because of his failure to feel appropriate

appreciation from his circle of acquaintances (Ahn et al., 2015; Alshehri & Elamrousy, 2023).

With the emergence of electronic exposure crimes due to fear of extortion or defamation of females, research has emerged as a reaction. Electronic friendships and the manipulation of partners' images, especially females, have become a subject of interest. In a study conducted by Wang & Wang (2010) on 800 online profile data, focusing on physical appearance, education level, occupation, personality, and interests, the results showed that females are more susceptible to having their pictures shared, which makes them more popular. Often, the feeling of psychological security in virtual reality arises due to the presence of three socially undesirable personalities known as the Dark Triad (Jones & Figueredo, 2013). The traits of the Dark Triad range from Machiavellianism, narcissism, to psychopathy. The threat to the victim's dominance arises from the perception of the victim's superiority or the selfishness to serve personal interests (Alavi et al., 2023). Non-compliance with the bully's requests represents a form of cruelty and manipulation that the victim suffers from, making them psychologically threatened, stigmatized, and humiliated (Brown et al., 2019). Manipulation cases of males online are often observed, as females tend to be more secretive and unwilling to disclose information to strangers or accept unfamiliar friendships, which helps maintain psychological security within cyberspace (Jahani & Eghbali, 2023). A study by Jahani and Eghbali (2023) finds that impulsivity, moral disengagement, and dark personality traits play a role in threatening online psychological security for teenagers.

Cao et al. (1993) reviewed the Psychological Security-Insecurity (SI) questionnaire developed by Maslow and examined its reliability and validity. Data were collected from 1,893 middle school students using the original SI-I questionnaire. The revised SI-I questionnaire consisted of 44 items, including 10 first-order factors and 3 second-order factors. It had good test-retest reliability, homogeneity reliability, and criterion validity, and is suitable for assessing the psychological security and insecurity of Chinese primary school students.

Nosek et al. (2002) performed a study that monitored the differences between traditional laboratory research and Internet-based research and their impact on the psychological security of both the researcher and the respondents. These differences had implications for research ethics (e.g., absence of the researcher relying on applications of artificial intelligence, potential exposure of confidential and/or identity data to third parties such as electronic application cases, ensuring debriefing and access to study tools produced by the researcher as he relies on a generative tool of artificial intelligence) and security (e.g., confidentiality and anonymity, security of data transmission, security of data storage, tracking of participants over time). We also review key design issues that a researcher must consider before implementing an Internet study, including the problem of self-selection of participants with the nature of the research, and the loss of experimental control over the Internet application. The study identified some challenges for Internet-based research, including the increased possibility of participants' misbehavior, whether intentionally or otherwise.

Wang et al. (2019) conducted a study using grounded theory on 25 urban areas and came up with three dimensions of qualitative analysis, after which they conducted a questionnaire and applied it to 802 participants. The scale consists of 20 items. Exploratory and confirmatory factor analyses were used to verify the scale's structure. The study found three dimensions: self-security, social environmental security, and natural environmental security. A highly reliable structure was obtained, and two items were excluded. Exploratory factor analysis reached four factors that explained 52.5% of the total variance of the phenomenon, and the fourth dimension was social security. Cronbach's alpha reliability for the scale was 0.77, and reliability coefficients ranged between 0.67 and 0.83.

Yilmaz and Aydin (2023) developed a measure of perceived job security among public primary school teachers. The sample consisted of 187 teachers. Exploratory factor analysis was used for the scale, which consists of 6 items. The items were divided into a general factor that explained 71.5%. Then, the study conducted a confirmatory factor analysis of the resulting structure on another sample of 211 teachers in Amasya.

The reliability achieved a high degree, and the overall reliability with the alpha coefficient for the scale reached 0.91.

The gap in studying the psychological safety of social media users appears to be due to a deficiency in the real-life environment that affects one's sense of self. Difficulties in adaptation have led individuals to adopt idealized virtual behaviors that differ from those exhibited in real-life interactions. Moreover, people's inclination to engage in conversations with strangers and their comfort with idealized interactions, which provide ideal feedback, have prompted them to disclose more information about themselves. At a certain point, individuals may feel exposed due to the intimate nature of the information they have revealed, and their personal or psychological security may be compromised if the other party discloses or exploits that information for malicious purposes. Individuals may become victims of cyberbullying, where their information, identity, and personal images are manipulated for unethical purposes. Consequently, the study proposes a scale for psychological security in the cyber realm to understand the threatened psychological processes of self and personality within a psychological framework. Thus, the study aims to answer the question: What are the underlying factors that govern the items of the psychological security scale in the cyber realm among university students in Omani society?

Methods

Sampling and sample profile

A purposive sample of 110 Omani university students was recruited from Dhofar University and Sultan Qaboos University, as these institutions represent major academic settings with diverse student populations relevant to the study objectives. The sample included 44 males (40%) and 66 females (60%). Participants were also categorized according to their daily social media usage: 58 students (52.7%) reported using social media for less than three hours per day, whereas 52 students (47.3%) reported usage exceeding three hours. All 110 responses were complete, with no missing data across any of the scale items. Participants provided oral consent in

accordance with ethical research procedures and were informed of their right to withdraw from the study at any stage without consequences (Moussa & Elnersh, 2025).

Procedures: The participants were tested on a touch testing application. They completed the scale responses of psychological security in cyberspaces. The sample participated in the measurement online at the Google Form website. The application duration was from October 2023 to February 2024.

Psychological security (PS) scale: The PS was measured by 42 items, which suggested common behaviors of social networking site users. In constructing and selecting items, the study relied on previous studies on psychological security and the characteristics of virtual reality, combining the characteristics of psychological security, virtual reality, and the symptoms of the ego. The study used studies (Al Diyar & Salem, 2015; Al-Mekaimi, 2023; Alshehri & Elamrousy, 2023; Cao et al., 1993; Molodetska et al., 2020; Nosek et al., 2002; Perlin & Li, 2020; Wang & Wang, 2010; Wang et al., 2019). The study identified three dimensions, considering the study by Cao et al. (1993), and seven general dimensions in light of blending the self in virtual reality.

The Emotional well-being subscale consisted of items 1 to 6, while the Personal Independence subscale consisted of items 7 to 12, the Confidence subscale consisted of items 13 to 18, the Credibility subscale consisted of items 19 to 24, and the Privacy and anonymity subscale consisted of items 25 to 30, the social support subscale is from items 31 to 36, and the Sense of power subscale is from items 37 to 42. The 5-point Likert scale has been used to respond to the items. Each item in the scale took scores from 1= never to 5= always. The items were written in the Arabic form.

Statistical analysis

The Jamovi 2.5.26 software is used for descriptive and multivariate data analysis. Descriptive statistics were conducted on subscales. Structural validity was performed by confirmatory factor analysis based on the maximum likelihood method. Evaluation of model fit depended on a set of indices: 1) X^2 value, CFI, TLI, NNFI is upper than .90, RMSEA is located within sequential limits of .05 and .08, and SRMR implies zero.

Results

Internal Consistency Reliability

Reliability was evaluated using Cronbach's alpha reliability coefficient and the omega coefficient for each of the subscales of the Psychological Security in Cyberspace scale. The results are shown in Table 1.

Table 1. Reliability coefficients and weighted means for the study subscales

Subscales	Alpha	Omega	Weighted mean	SE
Emotional well-being	.749	.772	3.23	.74
Personal independence	.820	.827	3.58	.72
Confidence	.925	.927	3.30	.88
Credibility	.890	.893	3.57	.75
Privacy and anonymity	.710	.731	3.57	.65
Social support	.949	.950	3.28	.95
A sense of power	.910	.914	2.95	.91

As shown in Table 1, It became clear from the weighted average values of the dimensions that they tend to agree in the first six dimensions, as they exceeded neutrality, and the minimum weighted mean score reached 3.23, and its maximum was 3.58 in the personal independence dimension, as the person is independent of the original reality in which he lives to acquire psychological support in a way that leads to psychological discharge. The results revealed acceptable stability in light of the omega coefficients, which ranged between values of 0.731 and 0.950.

Descriptive statistics:

Table 2 presents some descriptive indices of psychological security in cyberspace, as well as mean, sample size, minimum and maximum scores, skewness, and kurtosis. The results are as follows:

Table 2. Descriptive indices between study subscales.

	Emotional well-being	Personal independence	Confidence	Credibility	Privacy and anonymity	Social support	A sense of power
N	110	110	110	110	110	110	110
Mean	19.4	21.5	19.8	21.4	21.4	19.7	17.7
SD	4.45	4.34	5.26	4.48	3.92	5.64	5.58
Min	6	6	6	6	6	6	6
Max	30	30	30	30	30	30	30
Skewness	-.492	-.762	-.245	-.627	-.525	-.249	.241
Kurtosis	.342	1.84	-.297	.791	1.85	-.500	-.313

As shown in Table 2, the subscales scores ranged from 6 scores as a minimum score and 30 scores as a maximum score for each component. The skewness index ranged from .245 to .762 as an absolute value, which means the data was a normal distribution. 110 participants participated in the study, with no missing data found. The feel of credibility, privacy, and anonymity was higher in another component. Confidence in treatment with close and stranger friends has a value approximately with the median. Social support has an 19.7 average value. It was close to the median, which means the Omani people have good well-being in SNSs, and psychological security has an acceptable level.

The psychological dimensions identified through the subscales offer valuable insights into the state of psychological security among Omani university students in the digital realm. The mean scores reveal generally moderate to high levels of emotional well-being, personal independence, confidence, credibility, privacy, anonymity, and social support. However, the comparatively lower mean for the "sense of power" subscale suggests a potential area for growth in students' perceived digital empowerment. The skewness and kurtosis values indicate a slight leaning toward higher scores across most subscales, reflecting a predominant sense of psychological security within the sample. Notably, the elevated mean scores for personal

independence and credibility subscales underscore the cultural emphasis on self-reliance and trustworthiness, aligning with the societal norms of individual accountability and social cohesion in Oman. The variability observed in the social support subscale highlights differing perceptions of online networks' emotional and relational benefits. These findings illuminate the intricate interplay between individual psychological traits and cultural contexts, emphasizing the need to cultivate digital environments that enhance personal empowerment and community-oriented support systems for holistic psychological security.

Examination of structural validity

Exploratory factor analysis was conducted to determine the coherence of subscales. The principal axis factoring (PAF) method was tested, and Promax rotation was used for the correlation matrix. The study chose the value 0.4 as a cut-off score to accept the item factor loadings (Alharbi et al., 2023; Alenezi et al., 2024). Several factors were left free to obtain items on it. The items were extracted into three theoretical factors. Table 3 shows the item loadings on the three-factor model.

Table 3. Item loadings extracted on the three-factor model

Observed	Factors		
	1	2	3
A1	.73		
A2	.92		
A3	.88		
A4		.44	
A5	.85		
A6		.52	
A7	--	--	--
A8	--	--	--
A9	.68		
A10	.76		
A11	.81		
A12		.54	

Observed	Factors		
	1	2	3
A13	.80		
A14	.69		
A15	.76		
A16	.68		
A17	.62		
A18	.62		
A19			.57
A20	.73		
A21	.59		
A22	.46		
A23			.50
A24			.41
A25			.84
A26			1.01
A27		.41	
A28			.50
A29			.42
A30	.70		
A31	.75		
A32	.67		
A33	.54		
A34	.78		
A35	.75		
A36	.82		
A37	.55		
A38		.78	
A39		.79	
A40		.72	
A41	.56		
A42		.80	

As the results in Table 3, the factor analysis revealed three factors with eigenvalues of 15.29, 5.22, and 3.68, and explained 36.41%, 12.42%, and 8.77%, respectively. The KMO criteria were equal to .87. The results showed that the data were loaded in a parsimonious model in which each item was loaded in only one component. The data confirmed the three theoretical factors as Cao et al. (1993), but the study produced exploratory data to test the tetrad components only.

The problem of Cao et al. (1993) is that the model couldn't be found because the study produced a scale for Omani society. Then, EFA was conducted to test the second order by using the 7 factors as analysis units and performing the analysis by choosing 3 factors to be extracted. Principal component method used without rotation. The cut score of item factor loadings was .3. The KMO was .896, Bartlett's X^2 701, $P \leq .001$, and the total variance explained was equal to 85.2%. The interpretative nature of the scale was high. The seven factors were loaded on the general factor as shown in Table 4.

Table 4. The principal component factor analysis of the 7th factor.

Subscales	factor		
	1	2	3
Emotional well-being	.779		
Personal independence	.876		
Confidence	.906		
Credibility	.879		
Privacy and anonymity		.957	
Social support	.768		
Sense of power			.894

The existence of Cao et al. (1993) can't be found. The seven factors of psychological security have a general factor model.

The results highlight several critical aspects of the factor structure of the psychological security scale within Omani society, particularly through the application of exploratory factor analysis (EFA). Notably, the absence of the model from Cao et al.

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(1993) indicates that the original framework may not fully encompass the cultural and contextual nuances of psychological security in Oman. This underscores the limitations of directly applying Western-developed models to non-Western contexts without thorough cultural adaptation and validation.

The strong KMO value (.896) and the significant Bartlett's Test of Sphericity ($p \leq .001$) suggest that the data are appropriate for factor analysis, with the factors being sufficiently correlated to support the analysis. Moreover, the total variance explained (85.2%) is impressive, indicating a robust factor structure and reinforcing the scale's validity in assessing psychological security within this population.

The factor loadings for the seven subscales, emotional well-being, personal independence, confidence, credibility, privacy and anonymity, social support, and sense of power—demonstrate strong correlations with the general factor of psychological security. These high loadings (ranging from .768 to .957) suggest that each dimension significantly contributes to the overall construct, emphasizing that these factors are crucial for understanding the psychological experiences of individuals in Omani cyberspaces.

From a psychological standpoint, these results suggest that psychological security in cyberspaces for Omanis is a complex, multidimensional construct. Emotional well-being, confidence, and privacy emerge as particularly influential factors. At the same time, personal independence, credibility, and social support further highlight that psychological security is shaped not only by individual experiences but also by broader social and relational dynamics. This aligns with theories of social support and collective identity, suggesting that a sense of security in the digital world is influenced both by personal emotional resources and by one's social network and environment.

However, the inability to replicate the Cao et al. (1993) model signals the need for future research to delve into the unique sociocultural factors affecting psychological security in Oman. Specifically, exploring how local norms and values shape perceptions of privacy, social support, and power dynamics in digital interactions is essential. As

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Oman continues to evolve culturally and technologically, there is a clear need to refine and expand the scale to better reflect these influences and capture the realities of Omani digital users. Furthermore, while the overall model demonstrates substantial explanatory power, validating it with diverse samples and conducting longitudinal studies will further strengthen these findings, enhancing the scale's applicability across various settings.

The competition models of the factorial model for psychological security structure

Confirmatory factor analysis is tested for the seventh and three-factor models. Table 4 shows the user model versus baseline model goodness-of-fit as indicated.

Table 4. Goodness-of-fit of competition models for the psychological security scale

Model	X ² (df)	CFI	TLI	RMSEA	SRMR	GFI
CFA Seventh factor	2056 (798)*	.984	.983	.120	.090	.978
EFA Three-factor	1185 (491)*	.775	.758	.113	.826	.776

Note: * significance at 0.05, CFA refers to confirmatory factor analysis, and EFA refers to exploratory factor analysis.

Table 4 shows that the seventh-factor model is highly fitted according to CFI, TLI, SRMR, and GFI indices more increased than the three-factor model. The chi-square index and RMSEA are poorly fitted because of the nature of Omani youth which has a moral usage of social networking sites, and more applications are not working in the Sultanate of Oman, like TikTok, Telegram, Facebook messengers, calling ... etc. Another reason for these poor indexes' fitting is to social desirability of youth or attitudes toward positive orientation quality of life. Otherwise, the third reason is the small number of participants who responded to the scale item. Table 5 shows the factor item loadings.

The comparison between the seventh factor and three-factor models provides valuable insights into the validity of the psychological security scale in the Omani context. The seventh-factor model outperforms the three-factor model across several

key fit indices, including CFI, TLI, SRMR, and GFI, all of which suggest a stronger model fit. Specifically, the seventh-factor model demonstrates excellent fit, with CFI (.984) and TLI (.983) significantly higher than the three-factor model's values of .775 and .758. However, both models show poor fit in terms of RMSEA and chi-square indices, indicating challenges in adapting the model to the data. This discrepancy may be due to cultural and social factors unique to Omani youth, such as the moral use of social networking sites, social desirability bias, and limited use of certain digital platforms. Additionally, the small sample size could have contributed to the poor fit indices, particularly affecting chi-square and RMSEA values.

Despite these challenges, the seventh-factor model is considered a more accurate reflection of psychological security in Oman, supported by the higher fit indices. This finding underscores the importance of considering cultural factors when applying psychological scales, as local norms and behaviors can shape the way psychological constructs like security are experienced. Future research should refine the model to account for these sociocultural influences, examining how factors like privacy, social support, and power dynamics are understood within the Omani digital context. Moreover, addressing the limitations of sample size and minimizing social desirability bias will be crucial for enhancing the model's robustness and ensuring its applicability in diverse settings.

Table 5. item factor loadings of the seventh-factor model of the psychological security scale

t	Observed	Estimate	SE	Z	P
Emotional well-being	A1	.79	.036	21.81	.000
	A2	.78	.036	21.96	.000
	A3	.74	.050	14.84	.000
	A4	.21	.080	2.68	.007
	A5	.82	.036	22.70	.000
	A6	.54	.066	8.10	.000

t	Observed	Estimate	SE	Z	P
Personal independence	A7	.50	.057	8.77	.000
	A8	.59	.052	11.39	.000
	A9	.73	.046	15.65	.000
	A10	.84	.026	32.11	.000
	A11	.82	.031	26.57	.000
	A12	.80	.037	21.55	.000
Confidence	A13	.89	.021	42.12	.000
	A14	.92	.018	51.99	.000
	A15	.94	.015	64.19	.000
	A16	.89	.023	39.37	.000
	A17	.78	.039	19.94	.000
	A18	.82	.033	24.73	.000
Credibility	A19	.61	.061	10.07	.000
	A20	.92	.022	41.35	.000
	A21	.87	.027	32.34	.000
	A22	.78	.037	20.95	.000
	A23	.82	.029	28.60	.000
	A24	.85	.027	31.40	.000
Privacy and anonymity	A25	.23	.061	3.71	.000
	A26	.27	.053	5.08	.000
	A27	.82	.047	17.49	.000
	A28	.47	.059	8.01	.000
	A29	.40	.064	6.23	.000
	A30	.95	.034	27.95	.000
Social support	A31	.94	.013	71.25	.000
	A32	.91	.021	44.24	.000
	A33	.85	.024	34.70	.000

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t	Observed	Estimate	SE	Z	P
	A34	.94	.014	69.28	.000
	A35	.91	.019	49.16	.000
	A36	.93	.016	58.89	.000
Sense of power	A37	.89	.027	32.70	.000
	A38	.73	.040	18.17	.000
	A39	.85	.026	32.18	.000
	A40	.92	.019	48.61	.000
	A41	.94	.020	46.81	.000
	A42	.72	.049	14.48	.000

The social support loadings ranged between 0.85 and 0.94, which is high, so perceived social support is a path to achieving psychological security. The Sense of Power loadings ranged between 0.72 and 0.94, which is medium to high. A person's feeling of power stems from emotional release or social compensation for the shortcomings in his real-life interactions, and from the amount of sympathy he finds from his circle of acquaintances and friends through social media platforms. The item loadings of the Privacy and Anonymity dimension ranged between 0.23 and 0.95, and the dimension includes some weak loadings, so some individuals engage in cultural communication with anonymous friends for psychological relief. The loadings of the Emotional well-being dimension ranged between 0.21 and 0.82, and most of the item loadings are weak, which means a decline in emotional well-being as an indicator of psychological security in the cyber reality. Loadings for the Personal Independence dimension ranged between 0.50 and 0.84, and most loadings were moderate.

Correlation matrix of study variable

Table 6 shows the relationships of study variables using the Pearson correlation matrix. Demographic variables were gender and SNS time usage by hours, and the continuous variables were the psychological subscales and overall score. The results are as follows:

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Table 6. Correlation matrix between study variables

	Subscales	1	2	3	4	5	6	7	8	9
1	Emotional well-being	--								
2	Personal independence	.772* **	--							
3	Confidence	.730* **	.800* **	--						
4	Credibility	.676* **	.778* **	.802* **	--					
5	Privacy and anonymity	.522* **	.635* **	.576* **	.727* **	--				
6	Social support	.752* **	.787* **	.842* **	.779* **	.623* **	--			
7	Sense of power	.682* **	.680* **	.741* **	.581* **	.547* **	.775* **	--		
8	Psychological security	.847* **	.896* **	.914* **	.875* **	.748* **	.927* **	.841* **	--	
9	Gender	-.069	-.133	.259* *	-.202*	-.079	-.149	-.109	-.168	--
10	SNSs time usage	.126	.052	-.028	.034	.052	.061	-.046	.037	.178

Note. * $P < .05$, ** $P < .01$, *** $P < .001$

Gender negatively affects credibility, meaning that males have more credibility on social networking sites. The association had so small values. The lower values of association between confidence and gender mean that males use the SNSs in chats, posts, news, fashion, and market discounts. Otherwise, males or females use the SNSs with close friends.

Ego strength plays a role in the absence of differences between the sexes on the sub-dimensions of the psychological security scale. In addition to this, ideal and good relationships and realistic interactions that address the imbalance of reality were the cause of feelings of psychological well-being, and personal independence is not significant with the gender variable. Emotional release for both genders is the reason

for perceived well-being from supportive advice and guidance, and engaging in more balanced friendships that support psychological safety is the result.

It was noted from the results in Table 6 that the number of hours of using social networking sites did not affect psychological security or its sub-measures. This means that a person may necessarily be behaviorally addicted to using social networking sites, but they do not feel psychologically secure because of this use. A person may have reached the point where he has reached dependence, where he produces weak content that does not make him feel psychologically secure, as the content displayed is linked to obtaining admiration from people he does not know, and this gives the person inauthentic personality traits. Other people may produce purposeful content and produce material that is circulating and useful to its viewers, and thus this gives a person the conviction of temporary psychological security, but he quickly feels anxious for fear of being trolled or hacked to obtain his channel through social networking sites, as indicated by the study of Fakhrou et al. (2022) that cybercrimes are sometimes motivated by jealousy and attack.

Discussion

The objectives of this study were to build and validate a measure of psychological security in cyber reality among undergraduate students in the Sultanate of Oman. The results showed adequate psychometric properties, including good consistency, test reliability, and internal validity with the seven-factor model, while the three-factor model had a poor fit for Omani university students.

The study proved the existence of three theoretical factors around which vocabulary is organized, using exploratory factor analysis, but by studying the empirical meaning of the distribution of this vocabulary among the factors, it became clear that there is no explanatory significance for this distribution. Although this analysis agreed with the three-factor structure of the scale of Cao et al. (1993), in the current study, he did not adopt Maslow's theory of identifying needs, considering that psychological security in cyber reality amounts to a quest for ideal perfection to achieve a state of compensation for the problems of real reality. In addition, Kao's scale was

based on ten factors in the real environment, while in the current study psychological security problems during virtual online interactions were based on seven factors that differ in form and content. Omani society also maintains a unique nature in virtual interactions, and its interactions are limited to the narrowest limits, except in the case of close friends.

Also, a person's traits are matched by the opposite in personality, meaning that an introverted person in real life is matched by extroverted personality traits in virtual reality. This is due to the nature of the feedback he receives during psychological and emotional discharge processes, which raises states of happiness and psychological well-being, thus increasing states of psychological security during interactions. Virtual Reality. This is partly consistent with the study of Perlin & Li (2020); Shin (2010); and Wang (2010).

The sense of power that appears through electronic space seems insignificant considering the weighted average index, as virtual relationships and interactions are false interactions based on modifying the current situation into a reality that satisfies its owner, and interactions may confirm the causes of problems that occur in the individual's living environment. Accordingly, a person feels a lack of psychological security, in several forms, including feeling naked in front of his friends who know his problems, a feeling of low self-esteem as he seems unable to solve his problems, his expectation that the other party will reveal his secrets to the public, and his falling as a victim, as is reported in the media. Blackmailing victims from acquaintances and strangers. He felt guilty for revealing the smallest details, which showed him unable to deal with his problems.

Through the correlation coefficients, it became clear that the relationship between the number of hours of use of social networking sites and psychological security is not significant, as the increase in working hours via the Internet may be due to the motivation for entering the Internet, it may be for entertainment, or knowledge, or following the news, or for wasting free time, and this It is not related to psychological security, but rather it may be a defensive ploy to escape from the problems of the current

reality with its details and dilemmas. The consuming desire to interact with friends may be to justify one's actions, which increases the feeling of psychological insecurity, or the interaction via the Internet may be to fulfill work requirements, and thus the motive for security and psychological safety is completely unjustified.

Psychological security may be linked to the nature of the educational problems that the learner suffers from, as the learner resorts to artificial intelligence, whether to psychological counseling programs, Medpot GPT, as a means of psychological relief, and thus the person becomes one with the machine and suffers from psychological disorders because of their feeling of alienation and lack of trust in others. The period of use measured by time across artificial intelligence platforms may lead to completing educational assignments, which is a sign of a lack of trust and credibility because of his feeling of guilt that he resorted to ready-made templates of knowledge produced by artificial intelligence.

The study did not rely on studying the effect of demographic variables on the study variables in the sub-scales, in order not to reach biased results, especially in the gender variable, as females tend to seek social approval for some characteristics related to psychological security, and they also believe that with this data, they may become a victim of disclosure of survey results.

Conclusion

The structural validity of psychological constructs is vital for creating assessment tools that are both reliable and culturally appropriate. Psychological security, as a multi-dimensional construct, varies significantly across cultural contexts due to differences in societal norms, values, and technological advancements. While previous studies, such as Cao et al. (1993) and AboZaid (2023), provided theoretical frameworks for understanding psychological security, their applicability to non-Western contexts, particularly Oman, remains underexplored. The unique sociocultural dynamics of Oman, which blend traditional values with modern influences, necessitate a tailored approach to ensure the construct's relevance and validity.

This study addressed this gap by comparing two factorial models of psychological security: a three-factor model grounded in theoretical constructs and a seven-factor model tailored to Oman's cultural context. Exploratory and confirmatory factor analyses were employed to evaluate these models' structural coherence and cultural fit. The three-factor model, derived through exploratory factor analysis (EFA) using the principal axis factoring method with Promax rotation, identified three dimensions—emotional well-being, personal independence, and privacy—with eigenvalues of 15.29, 5.22, and 3.68, explaining 36.41%, 12.42%, and 8.77% of the variance, respectively. While this model demonstrated structural parsimony and alignment with theoretical foundations, it lacked the depth to fully capture the nuanced psychological and cultural dynamics of Omani youth.

The seven-factor model, specifically developed to reflect Oman's unique sociocultural context, demonstrated superior goodness-of-fit indices through confirmatory factor analysis (CFA), with CFI = .984, TLI = .983, SRMR = .090, and GFI = .978, outperforming the three-factor model (CFI = .775, TLI = .758, SRMR = .826, and GFI = .776). Although the chi-square (χ^2) values and RMSEA for both models were less optimal, these results are likely influenced by cultural factors such as digital behaviors, social desirability biases, and the study's limited sample size. The seven-factor model provides a more comprehensive framework, capturing the complexity of psychological security in the digital age and offering greater cultural relevance for the Omani context.

While the three-factor model serves as a foundational understanding of psychological security, the seven-factor model addresses the nuanced dimensions critical for effective assessment in Oman. These findings highlight the need for culturally tailored interventions to strengthen dimensions such as emotional well-being and privacy. Future research should involve larger and more diverse samples to enhance model generalizability across different populations. Additionally, addressing cultural influences on digital behaviors—such as promoting positive online interactions and



safeguarding personal privacy—could further strengthen psychological security among Omani youth.

By refining the structural understanding of psychological security and offering a culturally sensitive framework, this study contributes to the broader discourse on culturally adaptive psychological assessment practices. It underscores the importance of localized validation processes in advancing psychological research and practice.

Recommendations

To enhance the Omani scale's ability to measure psychological security in cyberspaces, several strategic modifications are needed to better capture Oman's unique cultural and technological landscape. First, it is essential to expand the scale by incorporating dimensions that address the specific challenges faced by Omani youth in digital environments. Psychological security in cyberspaces involves more than emotional well-being and privacy; it must also account for factors such as digital identity, online social interactions, and perceptions of digital safety. With Oman undergoing rapid digital transformation and social media usage on the rise, it is crucial to include components like trust in digital platforms, the impact of cyberbullying, and awareness of online privacy risks. These factors significantly affect the psychological well-being of individuals online and should be reflected in the updated scale.

Additionally, refining the scale's language to incorporate culturally relevant terminology and real-life examples will increase its contextual validity. Collaboration with local experts, including psychologists, sociologists, and youth representatives, can ensure that the tool accurately captures the experiences and concerns of Omani youth in the digital era. In Oman, familial and societal values have a significant impact on individual behaviors. Therefore, the scale should consider how cultural factors such as parental control over internet use, peer pressure in digital spaces, and societal expectations about online conduct influence psychological security. By addressing these cultural nuances, the scale can provide a more holistic and culturally sensitive assessment of psychological security in Oman.

Limitations and Directions for Future Research

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While this study provides valuable insights into the structural validity of psychological security in Oman, several limitations warrant attention in future research. One major limitation is the sample size and its lack of diversity. Although the study's findings are meaningful, they are derived from a small, homogeneous sample that may not fully capture the diverse experiences across different demographic groups, such as age, gender, or socioeconomic status. Future research should include a larger and more diverse sample to improve the generalizability of the results. Additionally, expanding the sample to encompass participants from different regions of Oman would provide a more comprehensive understanding of how psychological security varies across the country's diverse cultural and geographical contexts.

Another limitation is the focus on just two factorial models of psychological security. While these models offer valuable insights, they may not fully capture the multidimensional nature of psychological security in cyberspaces. Future studies could explore alternative models or adopt more qualitative approaches, such as in-depth interviews or focus groups, to gather richer, more nuanced data. These qualitative methods could uncover additional dimensions of psychological security that quantitative approaches may overlook, particularly in a culturally unique context like Oman.

Finally, the reliance on self-report measures in this study introduces potential biases, including social desirability and response biases, which could affect the accuracy of the findings. Future research should consider a multi-method approach, combining self-reports with behavioral observations, peer assessments, or physiological measures, to offer a more accurate and comprehensive understanding of psychological security. Longitudinal studies could also provide valuable insights into how psychological security in cyberspaces evolves, particularly in response to the rapid pace of digital innovation and shifting online behaviors. These efforts will not only help refine the scale but also ensure its continued relevance for future research and practical applications in Oman.

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